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HABIT LOOP MIRROR

“Screen Time tells you how much. We tell you why.”

Understand the patterns behind your digital habits.

Machine Learning and Applied AI Internship 2026

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Project theme: AI-Powered Digital Habit Awareness & Behavioral Self-Regulation

THE GAP

The Problem: screen time tells us how much, but not why

Key insight: Measuring screen time alone does not explain the behavior behind it.

- **What tools show:** Traditional tools show total screen time, app duration, and session counts.
- **What they miss:** They rarely explain recurring patterns, preceding context, or intentionality.
- **What remains absent:** They do not suggest meaningful alternatives for unplanned usage.

Traditional tracker: Duration → **Habit Loop Mirror:** Pattern → Context → Reflection → Action

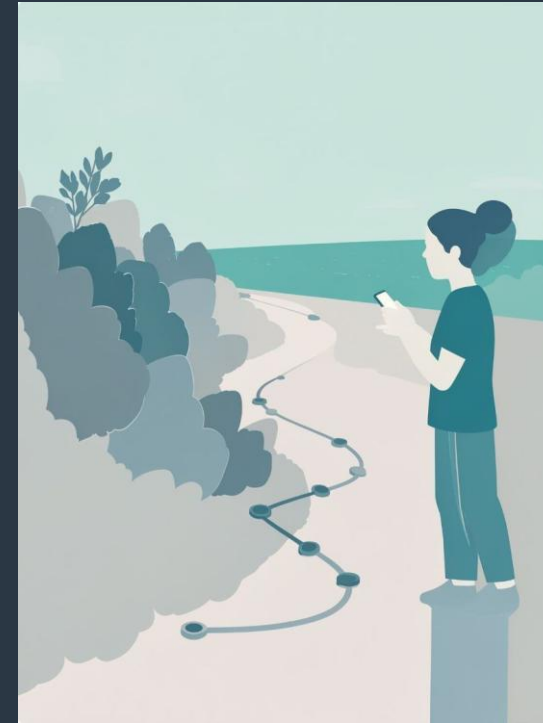


THE SHIFT

From screen-time metrics to behavioral awareness

Habit Loop Mirror transforms raw usage data into understandable patterns, evidence-based insights, reflection, and meaningful alternatives.

Understand → Reflect → Substitute → Improve



Understand	Reflect	Substitute	Improve
Detect recurring usage patterns.	Explain patterns using analytics and encourage non-judgmental reflection.	Suggest personalized alternatives.	Track multi-week habit evolution.

CORE MECHANISM

The 5-stage Behavioral Mirror Loop

Use evidence and non-judgmental behavioral language throughout.

Detect

Correlate notifications with session starts across multiple days.

Improve

Track multi-week evolution and verified reclaimed time.



Explain

Trigger → Action
→ Latency →
Duration.

Reflect

Planned,
Necessary,
Relaxation, or
Unplanned.

Substitute

Personal Swap
Engine
recommends 2-,
10-, or 20-minute
activities.

SYSTEM DESIGN

System Architecture



Layer	Technology
Frontend	React 18, TypeScript, Vite, Tailwind CSS, React Router, Lucide React, Framer Motion
Authentication	Supabase Auth + JWT
Backend	Python 3.12+, FastAPI, Uvicorn, SQLAlchemy, Pydantic, Alembic
Analytics	pandas, NumPy, scikit-learn
Intelligence	Deterministic Personal Swap Engine + Kimi K3 / Moonshot AI Adapter
Data	Supabase PostgreSQL; SQLite for local development
Deployment	Vercel → frontend; Render → backend

AI & Machine Learning intelligence

Python analytics generate the ground-truth evidence; AI explains that evidence.

ANALYTICS / ML

Evidence first

Processes telemetry, detects multi-day clusters, calculates latency and duration, and generates evidence-based signals.

AI LAYER

Meaning second

Converts evidence into understandable insights, contextual explanations, reflection, and personalized guidance.

FALLBACK

Continuity always

Deterministic evidence templates keep the system functional when Kimi access is unavailable.

Key features: Habit Loop Detection · Explainable Insights · Session Reflection · Personal Swap Engine
· Digital Day Timeline · Habit Evolution

How Habit Loop Mirror works

01 Start Habit Mirror	02 Sign up or enter Demo Mode
03 Complete personalization onboarding	04 Import usage CSV
05 Analyze usage data	06 Detect a recurring habit loop
07 View the explainable AI insight	08 Reflect on session intentionality
09 Select a personalized alternative	10 Track habit improvement

Example flow: Raw Usage Data → Recurring Evening Pattern → Evidence-Based Detection → AI Explanation → Reflection → Personal Swap → Long-Term Improvement

Responsible AI: awareness, not judgment



Awareness, not diagnosis

Not a medical treatment or clinical diagnostic platform.

Non-judgmental language

Avoid “addicted” labels and unsupported causal claims.

Evidence-based AI

Python analytics precede AI interpretation.

User control

Users choose categories and preferred alternatives.

Deterministic fallback

Core functionality remains available without external AI.

Self-regulation

Support awareness, reflection, and user-controlled improvement.

Avoid: “Notification caused the user to open the app.” **Use:** “Session followed notification.”

SDG 3 · GOOD HEALTH AND WELL-BEING

SDG 3: digital well-being is part of well-being

Habit Loop Mirror promotes digital well-being by helping users understand screen-time patterns and recurring digital habits, encouraging self-awareness, self-regulation, healthier routines, and meaningful alternatives to unplanned usage.



- Increased digital self-awareness
- Better understanding of recurring patterns
- Healthier digital routines
- Meaningful alternatives
- Long-term self-regulation
- Enhanced digital self-awareness
- Improved behavioral understanding
- Healthier and more intentional digital habits

READINESS & NEXT HORIZON

Validation, deployment & future scope

From measuring screen time to
understanding digital behavior.

THANK YOU

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Testing	Pytest backend tests · frontend production build and type checking · API health checks · deterministic demo dataset
Deployment	Vercel frontend · Render backend · Supabase PostgreSQL · Supabase Auth · Moonshot AI / Kimi
Future scope	real-time mobile telemetry · advanced pattern models · learned recommendations · real-time detection · privacy-preserving on-device analytics · expanded cross-platform support